

Programming Tools for Scientific Computing: Intermediate Project



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1 Problem Description

The *Brusselator* is a theoretical model used to study oscillatory behavior in chemical systems. It serves as a simplified representation of complex reactions such as the *Belousov-Zhabotinsky* reaction, which exhibits periodic changes in chemical concentrations over time.

In this model, four chemical species are involved: C_1 , C_2 , C_a , and C_b . To simplify the analysis, C_a and C_b are assumed to be present in such large quantities that their concentrations remain constant throughout the reaction. That is, $C_a = C_b = \text{const.}$. Additionally, all reaction rate coefficients are set to 1 to further simplify the system.

Under these assumptions, the dynamic behavior of the remaining chemicals, C_1 and C_2 , is described by the following nonlinear system of ordinary differential equations (ODEs):

$$\frac{dC_1}{dt} = C_a + C_1^2 C_2 - C_b C_1 - C_1 \quad (1)$$

$$\frac{dC_2}{dt} = C_b C_1 - C_1^2 C_2, \quad (2)$$

with the initial conditions for the solution variables

$$C_1(t=0) = C_2(t=0) = 1. \quad (3)$$

By grouping the solution variables into a state vector $\mathbf{C}$, the system of equations can be compactly written in vector form as $d\mathbf{C}/dt = f(\mathbf{C})$.

2 Numerical Solution

The system of ordinary differential equations (ODEs) will be solved numerically using two time-integration methods: (a) the explicit Euler method, which is first-order accurate, and (b) the classical Runge-Kutta method of fourth-order accuracy.

The explicit Euler method updates the state vector $\mathbf{C}_n$ at time step t_n according to the following formula

$$\mathbf{C}_{n+1} = \mathbf{C}_n + \Delta t f(\mathbf{C}_n), \quad (4)$$

with Δt being the time step size.

The explicit Runge-Kutta scheme updates the state vector with the following scheme

$$\mathbf{C}_{n+1} = \mathbf{C}_n + \frac{\Delta t}{6} (\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \quad (5)$$

with

$$\begin{aligned}\mathbf{k}_1 &= f(\mathbf{C}_n) \\ \mathbf{k}_2 &= f(\mathbf{C}_n + \Delta t \mathbf{k}_1/2) \\ \mathbf{k}_3 &= f(\mathbf{C}_n + \Delta t \mathbf{k}_2/2) \\ \mathbf{k}_4 &= f(\mathbf{C}_n + \Delta t \mathbf{k}_3).\end{aligned}$$

3 Task

1. Implement the explicit Euler method and the classical Runge-Kutta method as given above in Sec. 2 into the existing solver structure.
2. Create an output `.csv` using `std::ofstream` file and write all simulated time steps, t_i , and the concentrations C_1 and C_2 into the output file.
3. Choose a time step width of $\Delta t = 0.15$ s and simulate the system up to 30 s. Run the simulation with $C_a = 1$ and $C_b = 1.7$. Then change the parameters to $C_a = 1$ and $C_b = 3$. For each combination of parameters, run one simulation using the Runge-Kutta scheme, and one with the Euler method.
4. Prepare a python notebook, in which you load your simulated data and compare the results.
5. How does the system react to the changes in the parameters C_a and C_b ? How are the results influenced by the numerical method?